

Summary

Delving into the datasets, I've uncovered intriguing trends in Ecuadorian supermarket transactions, zooming in on specific stores, departments, and the various factors influencing supermarket activities. Notably, stores in **Quito, Pichincha**, take the lead in maximum **transactions**, emphasizing their significance in the dataset. However, **Store number 53 in Manta, Manabi**, stands out for its highest number of **on-promotions**, showcasing a proactive approach to promotional activities.

At the department level, **family**, distinctive performances came to light. **Manta and El Carmen, Store 53 and 54** in the Grocery 1 and Cleaning family, and Store 53 in the Beverages and Produce family, exhibited the highest on-promotion growth with the *"Transfer"* status under **type**. These insights provide valuable information for potential strategic planning, especially related to holidays. Similarly, in **Quito, Pichincha, Store 49** in the Produce family, categorized under the *Holiday* store type, demonstrated substantial on-promotion growth on three separate occasions coinciding with actual holidays.

Moving to the transaction and oil tables, they did yield insightful observations, too. The Daily oil prices, `dcoilwtico`, resulted in a wide range of records, with the lowest at 26.19 (West Texas Intermediate) and the highest at 110.62 (WTI). On *August 15, 2017*, multiple stores maintained a consistent oil price of 47.57 (WTI), indicating stability across various locations. Conversely, changes in daily oil prices are observed, starting on *January 3, 2013*, in **Quito, Pichincha**, with a value of 92.16 (WTI) and concluding on *August 15, 2017*, in **El Carmen**, at 47.57 (WTI), reflected fluctuations in oil prices over time and across different locations.

Further analysis of the earthquake on April 16th, 2016, revealed significant changes in average unit sales and transaction volumes. Notably, Store 44 in Quito, Pichincha, consistently led in transaction rates both before and after the earthquake, showcasing its remarkable resilience. Despite having the highest number of transactions in the dataset, Store 44 experienced a substantial 74% impact on its transactions, highlighting the challenges posed by the earthquake. These initial findings provide a glimpse into the nuanced dynamics of supermarket transactions, laying the groundwork for a more comprehensive understanding. While informative, the need for different approaches and further analysis to draw more conclusive and refined conclusions for informed decision-making will be explored.

Features

There are a total of six data sets but only four were analyzed for this process. I will focus on specific features/columns from the different data sets, which are as follows:

- 1. Store number
 - 2. Family
 - 3. Sales
 - 4. On promotions
 - 5. State
 - 6. Type
 - 7. Date
 - 8. Daily oil prices
 - 9. Transactions
 - 10. Transferred
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Tables

Train table

Store_Amount	Department_Amount	Sales_Amount	Sales_Total	On_Promotic
54	33	379610	1073644952.20309	362

Transactions' table

Store_Amount	Transactions_Amount	Transactions_Total	Earliest_Date	Latest_Date
54	4993	141478945	2013-01-01	2017-08-15

Stores' tables

Store_Amount	City_Amount	State_Amount	Type_Amount	Cluster_Amount	Cluster
54	22	16	5	17	458

Oil tables

Oil_Amount	Oil_Total	Earliest_Date	Latest_Date
999	79564.38000000001	2013-01-01	2017-08-31

Holidays_Events' table

Type_Amount	Locale_Amount	Locale_Name_Amount	Description_Amount	Transferr
6	3	24	103	2

